CSS Properties: A Simple Guide

A beginner-friendly explanation of every unique CSS property and concept in your card design.

1. Understanding the CSS selectors

body selects the page body. **.card** selects an element whose class is card. **h2**, **form**, **input** and **button** select those HTML elements. **button:hover** selects a button only while the mouse pointer is over it.

Remember: a dot (.) before a name selects a class. The colon in :hover introduces a pseudo-class (a special state of an element).

2. The body selector

```
body {\n font-family: 'Lucida Sans', 'Lucida Sans Regular',\n 'Lucida Grande', 'Lucida Sans Unicode', Geneva, Verdana, sans-serif;\n background-color: black;\n margin: 0;\n padding: 0;\n height: 100vh;\n display: flex;\n justify-content: center;\n align-items: center;\n}
```

font-family

Meaning: Chooses the font used for text.

```
font-family: Arial, sans-serif;
```

The browser tries the fonts from left to right. If one is unavailable, it tries the next. sans-serif is a generic fallback font family.

background-color

Meaning: Sets the background colour of an element.

```
background-color: black;
```

In your code, the page background is black. You can use colour names, hexadecimal values such as #ffffff, or other CSS colour formats.

margin

Meaning: Creates space outside an element.

```
margin: 0;
```

Your body has its default outside margin removed. This helps the page fill the viewport without the browser's usual body gap.

padding

Meaning: Creates space inside an element, between its content and its border.

```
padding: 0;
```

Your body has no inner padding. The .card later uses padding: 30px.

height

Meaning: Sets an element's height.

```
height: 100vh;
```

vh means viewport height. 100vh is the height of the browser's visible window. 50vh would be half of it.

display

Meaning: Controls how an element's contents are laid out.

```
display: flex;
```

This makes the body a flex container. Its direct child elements (such as the card) become flex items.

justify-content

Meaning: Aligns flex items along the main axis.

```
justify-content: center;
```

With the default flex-direction: row, the main axis is horizontal. center places the card in the horizontal centre.

align-items

Meaning: Aligns flex items along the cross axis.

```
align-items: center;
```

With the default row direction, the cross axis is vertical. center places the card in the vertical centre.

Why the card is centred

These three declarations work together: **display: flex** turns the body into a flex container, **justify-content: center** centres its child horizontally, and **align-items: center** centres it vertically. This explanation assumes the default flex-direction is row.

3. The .card selector

```
.card {\n background-color: aliceblue;\n border-radius: 8px;\n padding: 30px;\n width: 300px;\n}
```

border-radius

Meaning: Rounds the corners of an element.

```
border-radius: 8px;
```

A larger value makes the corners more rounded. 50% can make a circle when width and height are equal.

width

Meaning: Sets an element's width.

```
width: 300px;
```

px is a fixed CSS length unit. Important: with the default box-sizing: content-box, 300px is the content width. Adding 30px padding on each side makes the card's total width 360px (not including any border).

4. The h2 selector

```
h2 {\n color: #1a1a1a;\n text-align: center;\n margin-bottom: 20px;\n}
```

color

Meaning: Sets the colour of the text.

```
color: #1a1a1a;
```

#1a1a1a is a very dark grey. #fff is shorthand for #ffffff (white).

text-align

Meaning: Aligns inline text horizontally inside its block.

```
text-align: center;
```

Common values are left, right, center and justify.

margin-bottom

Meaning: Sets the outside space below an element.

```
margin-bottom: 20px;
```

This adds 20px below the h2 heading.

5. The form selector

```
form {\n display: flex;\n flex-direction: column;\n gap: 25px;\n}
```

flex-direction

Meaning: Sets the direction in which flex items are placed.

```
flex-direction: column;
```

row = left to right; row-reverse = right to left; column = top to bottom; column-reverse = bottom to top. Your form uses column so its direct children stack vertically.

gap

Meaning: Sets the space between flex items (or grid items).

```
gap: 25px;
```

This creates a 25px gap between neighbouring form flex items. It does not add space around the outside of the group.

6. The input selector

```
input {\n margin-bottom: 15px;\n padding: 6px;\n border-radius: 6px;\n border: 2px solid black;\n background-color: #d8d8d8;\n}
```

border

Meaning: Adds a border around an element.

```
border: 2px solid black;
```

This shorthand has three parts: **2px** is the thickness, **solid** is the line style, and **black** is the colour. Other styles include dashed, dotted, double and none.

The input also uses **margin-bottom: 15px** (space below each input), **padding: 6px** (inner space), **border-radius: 6px** (rounded corners) and **background-color: #d8d8d8** (light grey background).

Spacing note: Your form has gap: 25px and each input also has margin-bottom: 15px. If the inputs are direct flex items, both spaces can contribute to the distance between them. You may not need both.

7. The button and hover state

```
button {\n background-color: #1b81da;\n color: #fff;\n padding: 10px;\n border-radius: 8px;\n}\n\nbutton:hover {\n background-color: blue;\n}
```

:hover (pseudo-class)

Meaning: Applies styles while the mouse pointer is over an element.

Your button normally has a blue background (#1b81da). When the pointer moves over it, the background changes to blue. Other useful pseudo-classes include :active (while an element is being activated), :focus (when an element has focus), and :visited (a visited link).

8. Quick revision table

CSS property / concept	Simple meaning
font-family	Chooses the font.
background-color	Sets the background colour.
margin	Space outside an element.
padding	Space inside an element.
height	Sets height.
display	Controls layout; flex activates Flexbox.
justify-content	Aligns flex items on the main axis.
align-items	Aligns flex items on the cross axis.
border-radius	Rounds corners.
width	Sets width.
color	Sets text colour.
text-align	Aligns text horizontally.
margin-bottom	Sets outside space below an element.
flex-direction	Sets the direction of flex items.
gap	Space between flex or grid items.
border	Adds an element border.
:hover	Special state while the pointer is over an element.

9. Important things to remember

• **Margin** = outside spacing; **padding** = inside spacing.
• **color** changes text colour; **background-color** changes the background.
• Flexbox uses a **main axis** and a **cross axis**. With flex-direction: row, the main axis is horizontal; with column, it is vertical.
• You do not need to memorise every value. Practise changing values and observe what happens.
• CSS units used in your code: **px** = pixels; **vh** = viewport height.

10. What to learn next

A useful learning order is: CSS selectors → Box Model → width, height and units → Flexbox → CSS Grid → position (static, relative, absolute, fixed, sticky) → media queries → transitions and animations.

The CSS Box Model is the best next topic because it explains how content, padding, border and margin affect an element's size.